# Distilling an Ensemble into One Small Transformer: 2.10 → 1.41 BPB on WikiText-2

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**Abstract**

**Frozen final result (2026-09-29):** validation BPB **1.3947**, test BPB **1.4140** (baseline ≈ 2.10, **−0.69**), scored in FP32 on CPU. Evaluation budget: 31.61 s test CPU FP32 (4.14× the same-session baseline of 7.63 s, clean idle; limit 5×), 0.69 GiB peak RAM, 42.5 MiB inference assets (limits: 4 GiB, 64 MiB). Total training cost: ≈ 31.5 GPU-hours (RTX A2000 Laptop, BF16).

**Method in one line:** a nine-member logit ensemble ("super-teacher v3", validation 1.3840, training-time only) is distilled ($L = (1-\alpha)\,\mathrm{CE} + \alpha T^2 \mathrm{KL}$, $\alpha = 0.75$, $T = 3$) into a single 10.55M-parameter transformer with a modern architecture stack (RoPE, RMSNorm, SwiGLU, bias-free linears, zero-init output projections); the absorption gap is closed by low-lr WSD continuation; the frozen weights are then refined by global temperature calibration ($T = 1.08$), two short low-lr KD continuations and in-basin weight averaging; a post-freeze CPU inference fast path (fused attention and elementwise kernels, more intra-op threads — §1.7) brings scoring inside the time budget without touching weights. The final model scores **1.414029** test BPB in one forward pass, ending 0.011 above its ensemble teacher.

# 1 Method

## 1.1 Overview

The final solution is a four-stage pipeline. Every stage below is selected on validation only; the test split is touched once, after freezing (Section 4).

9-member logit ensemble — teacher v3, validation 1.3840 (training-time only)
↓ KD: $L = (1-\alpha)\,\mathrm{CE} + \alpha\,(T_{\mathrm{ens}}T)^2\,\mathrm{KL}$, $\alpha = 0.75$, $T = 3$, teacher frozen
fresh student — 288 × 10 × 9 modern transformer, SGDR-3 × 14,400 steps
↓ two WSD continuation passes at lr 0.004, initialized from EMA (validation 1.4036)
frozen EMA 0.999
↓ $T = 1.08$ calibration (1.4036 → 1.3977) · two 2,400-step continuations at lr 0.001
equal-weight average of the two branches — final checkpoint, one forward pass
→ validation 1.3947, test 1.414029

## 1.2 The student architecture

A single transformer of width 288× depth 10× 9 heads, vocabulary 2,048, context 256, tied embeddings — 10,549,153 parameters (10.55M). Four modern-architecture upgrades, each FLOPs-neutral at equal width×depth, are enabled as config flags over the supplied implementation (contract tests 7/7):

- **RoPE** (base 10,000) instead of learned positional embeddings;
- **RMSNorm** instead of LayerNorm;

- **SwiGLU** feed-forward with hidden width $8w/3$, keeping FFN FLOPs unchanged;
- **bias-free linear layers** with **zero-initialized output projections**.

A paired screen at 2,400 matched steps favored modern over the legacy geometry ($-0.012$ raw / $-0.023$ EMA); the advantage decays with full training ($-0.003$ after WSD), but the modern student absorbs its teacher more completely (gap 0.0006 vs. 0.0032 legacy). The model also keeps the **backout lambda** [8]: before the final prediction, a learned scalar $\lambda$ subtracts the mid-network residual stream from the final one ($x_{\text{final}} - \lambda \cdot x_{\text{mid}}$, $\lambda$ initialized to 0). The learned $\lambda \approx 0.4$–$0.5$ and the mechanism is worth $\approx 0.007$ BPB. Nothing here is visible to the fixed evaluator: the checkpoint is a standard student checkpoint, one forward pass at scoring.

### 1.3 The super-teacher v3

Nine single models are combined into one **logit-space ensemble** with per-member weights and a global temperature (both tuned on validation): five checkpoints from the project's distilled lineage, the two strongest single models (legacy and modern), one deliberately weak, decorrelated member (plain recipe, validation 1.5058, weight 3) for error independence, and one seed variant. The ensemble scores **1.3840 validation BPB** — far beyond any single model at this FLOP budget (the single-model frontier is $\approx 1.40$). It exists only at training time, so the evaluation budget does not constrain it; at scoring time it is never used. Its only job is to supply soft targets.

### 1.4 The distillation loss

The loss follows Hinton et al. [9]:

$$L = (1-\alpha)\,\mathrm{CE}(\text{student}, y) \;+\; \alpha\,(T_{\text{ens}}T)^2\,\mathrm{KL}\Big(\mathrm{softmax}\Big(\tfrac{t}{T_{\text{ens}}T}\Big) \;\Big\|\; \mathrm{softmax}\Big(\tfrac{s}{T_{\text{ens}}T}\Big)\Big), \qquad (1)$$

where $t/s$ are teacher/student logits; $T_{\text{ens}} = 1.05$ is the teacher's own ensemble temperature, read from the teacher checkpoint; $\alpha = 0.75$ and $T = 3$ were selected on a 2,400-step validation grid. The teacher is frozen and forward-propagated on training windows online — no pre-computed targets, no extra storage. Label smoothing is disabled in KD runs (the soft targets already encode smoothing). The hard-target anchor is necessary at this scale: $\alpha = 1$ (pure KL) collapses to $\approx 1.54$ BPB. The student (10.55M) exceeds any single teacher member in capacity, so this is born-again distillation [10]: the ensemble's distributional knowledge is transferred into the one model the evaluation budget can afford.

### 1.5 Training recipe and schedule

**Optimizer.** All hidden 2-D matrices (QKV, projections, FFN) are trained with **Muon** [1] — Nesterov momentum 0.95 followed by a 5-step Newton–Schulz orthogonalization of each update, learning rate 0.02. Embeddings, the tied head, norm gains and scalars use an auxiliary Adam (lr 0.02, $\beta = 0.9/0.95$, $\varepsilon = 10^{-10}$). Implementation: the official KellerJordan `muon.py` (MIT), unmodified. At equal settings Muon beats AdamW by $\approx 0.03$ BPB on the matched runs, and its noisier cycle endpoints are smoothed by EMA — the two components are complementary [1, 2, 3].

**Regularization.** Decoupled weight decay 0.07, dropout 0.05 (embeddings and both residual branches), gradient clipping 1.0, no label smoothing under KD.

**Fresh phase — SGDR.** 3 cosine-restart cycles $\times$ 4,800 steps = 14,400 steps (batch 32$\times$ context 256 = $\approx$118M processed targets, $\approx$57 epochs of the 2.08M-token training set), warmup 100 steps, minimum-lr ratio 0.1 [4]. The length is chosen where the training loss stops falling steeply (it still is at 9,600 steps); the 14,400-step chain beats the 9,600-step chain at every stage (0.002–0.004).

**Continuation phase — WSD.** From the student's EMA weights, re-run the distillation with the warmup–stable–decay schedule [11] at lr 0.004 for 4,800 steps (decay over the last 2,400), twice (the second pass from EMA 0.999). This phase closes the absorption gap, and the schedule choice is decisive: a fresh student absorbs little of a strong teacher in one pass, but a *restart* (SGDR at lr 0.02) escapes the basin the student has found (1.4210 → 1.4486), while WSD at lr 0.004 preserves it and keeps absorbing (1.4210 → 1.4110 → 1.4102). With teacher v3 the chain reads 1.4093 (fresh EMA) → 1.4036 (pass 1) → 1.4036 (pass 2).

**Weight averaging.** An exponential moving average of the parameters (decay 0.999, updated every step, FP32) is the checkpoint used for continuation and freezing; it removes the large end-of-cycle oscillations that restarts and Muon induce.

### 1.6 Final refinements on frozen weights

Three levers act after the long chain converges, adding no parameters and no inference cost beyond one scalar division:

**(i) Global temperature calibration.** The final logits are divided by a scalar $T$ before the softmax: the KD-trained logits are slightly over-sharp for the reported BPB metric, because KD and EMA optimize the training objective, not BPB. A validation sweep on the frozen weights selects $T = 1.08$: 1.403555 → **1.397650**.

**(ii) Short low-lr continuations.** From the frozen EMA, re-run the KD with the WSD schedule at lr 0.001 for 2,400 steps (≈0.34 GPU-hours each) — a $\mu$-scale refinement inside the basin rather than a macro move. Two KD temperature formulations (`legacy_shared`: teacher and student both divided by $C \cdot T$; `teacher_calibrated`: teacher divided by $C \cdot T$, student by $T$, KL scaled by $T^2$) and two seeds replicate the gain: 1.395561 / 1.396005 (at $T = 1.08$). The two formulations differ by 0.0001 — noise — so the gain is attributed to the common short continuation plus EMA, not to the formula.

**(iii) In-basin weight averaging.** The two 2,400-step branches share initialization and recipe family, so their weights remain in one basin; equal-weight averaging (a "checkpoint soup", still one forward pass at scoring) reduces branch noise: **1.394684**. The same operation across *independently seeded* full runs catastrophically fails (3.20 BPB), because permutation symmetries misalign the weight spaces [6]; shared initialization is what makes averaging safe.

### 1.7 Budget accounting

Scoring FLOPs scale as $12 \cdot \text{depth} \cdot \text{width}^2 + 2 \cdot \text{vocab} \cdot \text{width}$ per token. On the submission machine (CPU FP32, same session, same split, baseline re-measured):

Table 1: Budget accounting on the submission machine.

| Model | Params | Test CPU FP32 | Assets |
|---|---|---|---|
| baseline | 1.09M | 7.63 s (1.0×, clean idle) | 4.4 MiB |
| **final** | **10.55M** | **31.61 s (4.14× same-session baseline)** | **42.5 MiB** |

Peak scoring RAM is 0.69 GiB (limit 4 GiB), measured externally. The time budget is met by an inference fast path added *after* freezing: a fused numba kernel replaces RoPE, the causal softmax and the $p \cdot v$ product of every attention block with one blocked flash-style pass (scores never materialize beyond a $64 \times 64$ tile), silu-mul and RMSNorm become single-pass elementwise fusions, and intra-op threading rises from the scorer default of 4 to 12. The fast path engages only in CPU FP32 evaluation mode; every other input keeps the original PyTorch formulation, and all seven contract tests plus a frozen-checkpoint equivalence check (max $|\Delta \log p| \approx 6.6 \times 10^{-5}$, validation BPB shift $8 \times 10^{-10}$) pass. The frozen checkpoint scores the full test split in 31.61 s — 4.14× the same-session baseline of 7.63 s, inside the 5× limit — with weights untouched and

printed BPB unchanged. The CPU-time ratio is machine-state-dependent and is reported with its pairing conditions (§3.4). Mixture-of-experts was rejected: the binding constraint is FLOPs, not parameters, so sparse routing trades exactly the scarce resource.

# 2 Experiments

All development and selection used the validation split (CPU FP32 scoring). The test split was scored only after freezing (eight times; Section 4). Training data, tokenizer and evaluator are unchanged (hashes verified). The numbers below are therefore the evidence for the method of Section 1.

## 2.1 Initial baseline

The supplied baseline (1.09M-parameter GPT, width 128× depth 4, tied embeddings) scores ≈ 2.10 test BPB (code README). Our reproduction scores 2.0718 validation BPB (CPU FP32). Four diagnosed limitations motivate the method: (i) **under-training** — 9.83M processed targets ≈ 4.7 epochs, training loss still falling steeply at the end (4.32); (ii) **no regularization** — longer training overfits immediately; (iii) **a stale optimizer** — matrix-preconditioned optimizers (Muon, SOAP) are more sample-efficient on small models in particular [1, 2, 3]; (iv) **an under-used budget** — the baseline uses 1× of a 5× CPU-time allowance and 4.4 of the 64 MiB assets, so a much stronger model is affordable. At matched 9.83M targets, restarts alone beat the baseline (2.0632 vs. 2.0718) while regularization without restarts underfits (2.1315) — regularization pays off only once training is long enough (§3.5).

## 2.2 The chain: where each component's gain comes from

Table 2: Validation BPB at each stage of the final chain (CPU FP32).

| Stage | BPB |
|---|---|
| baseline, reproduced | 2.0718 |
| teacher v3 (9-member logit ensemble; training-time only) | 1.3840 |
| fresh KD student, SGDR-3 × 14,400, EMA | 1.4093 |
| + WSD continuation pass 1 (lr 0.004) | 1.4036 |
| + WSD continuation pass 2 (lr 0.004) | 1.4036 |
| + $T = 1.08$ calibration (frozen weights) | 1.3977 |
| + two 2,400-step continuations (lr 0.001), two seeds | 1.3956 / 1.3960 |
| + equal-weight average of the two branches | **1.3947** |

## 2.3 Matched-budget comparison and ablation of the key mechanism

The key mechanism is distillation. Its ablation is training the same architecture on the same token budget without the teacher:

- **KD vs. plain recipe, matched budget** (2,400-step screen, same architecture, all else equal): KD ($\alpha = 0.75$, $T = 3$) beats the plain recipe by $-0.080$ validation BPB. This rules out "longer training alone" as the explanation — at equal processed targets, the teacher is what makes the difference.
- **The hard-target anchor:** $\alpha = 1$ (pure KL) collapses to ≈1.54; $\alpha = 0.75$ keeps the data term. The 2,400-step $\alpha \times T$ grid peaked at $\alpha = 0.75$ (smaller $\alpha$ under-uses the teacher), and $T = 4$ beat $T = 3$ by only $-0.0011$ — the operating point is the grid optimum, not a default.

- **Schedule ablation (same teacher, same student init):** continuing with an SGDR restart at lr 0.02 degrades the student (1.4210 → 1.4486); continuing with WSD at lr 0.004 improves it (1.4210 → 1.4110 → 1.4102). The WSD schedule is itself a key mechanism; its ablation is the restart. An independent seed trained from zero under the same recipe reaches 1.4208 (seed spread ≈0.0001), so the 0.011 WSD gain is ∼100× the noise.

Table 3: Ablation of the final refinements (frozen weights, CPU FP32).

| Variant | val BPB |
|---|---|
| frozen EMA, $T = 1.00$ (no calibration) | 1.403555 |
| frozen EMA, $T = 1.08$ | 1.397650 |
| + best single 2,400-step branch | 1.395561 |
| + average of both 2,400-step branches | **1.394684** |

Temperature sweep on frozen weights: $T = 0.96, 1.00, 1.04, 1.06, 1.08, 1.10, 1.12 \rightarrow 1.4111$, 1.4036, 1.3992, 1.3981, **1.3977**, 1.3978, 1.3986. The optimum $T = 1.08$ is re-confirmed on the averaged candidate, and the official evaluator in CPU FP32 scores it **1.3946842**, identical to the GPU sweep ($\Delta < 10^{-7}$). The knob is pure calibration: $T > 1$ softens the sharper-than-data distribution the student inherited from its soft targets, and the plateau over $T \in [1.04, 1.12]$ (spread 0.0015) shows the gain is robust — at zero added scoring cost.

## 2.4 Frozen evaluation

Table 4: Frozen evaluation of the final candidate (2026-09-29).

| Metric | Value | Limit |
|---|---|---|
| **test BPB** | **1.414029** | lower is better (baseline ≈ 2.10) |
| CPU FP32 time | 31.61 s (4.14× baseline 7.63 s) | 5× |
| Peak RAM | 0.69 GiB | 4 GiB |
| Inference assets | 42.5 MiB | 64 MiB |

The seventh test disclosure scores the frozen candidate at 1.414029 (−0.0094 versus the fourth freeze, 1.423472); the validation-to-test gap is 0.0193, inside the historical 0.0165–0.0199 range. Two independent runs agree to full precision (1.4140290484849805). The eighth disclosure re-scores the *same checkpoint* (SHA-256 unchanged) under the inference fast path of §1.7: test BPB 1.4140290449 — identical at the reported precision — in 31.61 s CPU FP32, 4.14× the same-session baseline and inside the 5× limit.

## 2.5 What the comparisons establish

- **Sections 2.1 and 2.3:** the baseline's four limitations are real: restarts, regularization, Muon and scale each contribute at matched budget, and KD beats the plain recipe by −0.080 at equal processed targets.
- **Section 2.2:** the chain attributes the total −0.68 to its components: ≈ −0.66 from training + distillation + continuation (2.0718 → 1.4036), ≈ −0.006 from calibration, ≈ −0.002 from short continuation, ≈ −0.001 from averaging — the last three at zero added inference cost.
- **Section 2.4:** the frozen candidate generalizes within the historical validation-to-test range, and the final model ends 0.011 above an ensemble that is 9× costlier to run at scoring (one forward pass).

# 3 Why the method works — and where it stops

## 3.1 Soft targets are a data-dependent label smoother

The teacher's per-token predictive distribution encodes the ensemble's uncertainty over the next word, which raw one-hot labels — and uniform label smoothing — cannot. That is why KD with the hard-target anchor ($\alpha = 0.75$) beats both the plain recipe ($-0.080$ at matched budget) and pure-KL distillation (which drifts to $\approx$1.54 without the anchor), and why students may exceed any teacher member in capacity (born-again [10]): the transfer is distributional knowledge, not just soft labels at the data's own difficulty.

## 3.2 The teacher's advantage is only partially representable by one model

Teacher v3 (1.3840) is 0.023 better than the previous five-member teacher (1.4070), but at the same chain stage the student improved by only 0.0066 (1.4102 $\rightarrow$ 1.4036), i.e. $\approx$29% of the teacher's gain transferred. A nine-member ensemble's advantage is substantially *model averaging*, which no single model of this capacity can represent — so a better teacher stops being a better curriculum. A teacher rebuilt from near-frontier members (validation 1.3849, members all within the frontier) did not transfer better either: what fails to distill is the ensemble value itself, not its distance from the student. The binding ceiling at this FLOP budget is single-model capacity; the final 0.009 therefore came from inside the basin (calibration, short continuation, averaging) rather than from more teacher signal.

## 3.3 How the student is driven matters as much as the teacher's strength

One full pass at the normal learning rate transfers only part of the teacher's advantage; the rest is absorbed only if continuation preserves the basin. A restart at lr 0.02 escapes the basin (1.4210 $\rightarrow$ 1.4486, worse than the fresh student), while WSD at lr 0.004 keeps absorbing (1.4210 $\rightarrow$ 1.4102; the first pass closes $\approx$2/3 of the gap). The same principle governs the final refinement: lr 0.001 for 2,400 steps improves the frozen model (1.3977 $\rightarrow$ 1.3956/1.3960), whereas the same continuation at lr 0.004 or 0.02 was already shown to plateau or destroy the basin. Distillation gain is thus a function of *how* the student is driven, not only of teacher strength.

## 3.4 Trade-offs between prediction quality and computational cost

Training costs $\approx$31.5 GPU-hours (RTX A2000 Laptop, BF16; $\approx$2.4 h recipe/scale/optimizer screens, $\approx$9.5 h KD grids and distillations, $\approx$5 h teacher search and WSD, $\approx$13.5 h fourth-round chains and probes, $\approx$1 h fifth-round refinements), plus a search cost of $\approx$150 CPU FP32 validation scores, 21 KD grid points and $\sim$50 ensemble-weight points — the price of the teacher. At scoring the final model costs one forward pass: 31.61 s CPU FP32, 0.69 GiB, 42.5 MiB — 4.14$\times$ the same-session baseline of 7.63 s, inside the 5$\times$ limit. The CPU-time ratio needs care: it varies with machine state, because the tiny baseline benefits more from an idle turbo-boosted CPU than the compute-bound model, so the multiple is always reported as a same-session pairing with its conditions rather than as a bare number. On the reference Xeon (code README: baseline scoring 5.92 s), the model's absolute time is what the automated verification measures.

## 3.5 Measurement discipline

Two pitfalls were caught only because selection is done with CPU FP32 scoring: (a) GPU-FP32 and CPU-FP32 validation scores differed by up to 0.015 BPB for identical weights — all reported numbers are CPU; (b) the same "9,600 steps" means different cosine phases under different total steps, so cross-run comparisons are valid only when paired within a run.

### 3.6 Mechanisms that failed and bounded the design

(i) A within-window neural cache [5] — three variants (logit-space addition, corrected key–value alignment, faithful probability-space mixing), grid-searched on frozen weights — gained at best $-0.0007$ and probability-space mixing *hurt* ($-0.005$ at $\alpha = 0.05$): causal attention already performs in-window retrieval, so the cache's copy distribution is redundant and dilutes the model. (ii) Cross-seed weight averaging fails catastrophically (3.20 BPB): permutation symmetries misalign the weight spaces [6]; averaging is used only inside one basin (§1.6). (iii) Students sharing a teacher lineage make poor ensemble members (the teacher already contains the sibling's signal, so diversity vanishes), and same-capacity self-distillation saturates after one round — both symptoms of the same mechanism: what transfers is the teacher's distribution, and once it is learned there is nothing left to combine. These negatives set the design boundaries; the final method lives exactly inside them.

## 4 Compliance

- Only the supplied training text [7] was used for learning; all settings and checkpoints were selected on validation. The test split was scored eight times, each only after freezing: 1.5714 (AdamW ensemble), 1.5410 (Muon ensemble, later found to exceed the time budget), 1.5359 (backout ensemble), 1.439251 (born-again distilled), 1.426762 (super-teacher distilled), 1.423472 (modern, nine-member teacher), 1.414029 (final), 1.414029 (final re-scored under the inference fast path of §1.7; checkpoint, data, tokenizer and evaluator hashes unchanged). Each re-freeze followed from validation-only experiments or budget re-measurement; no decision used test numbers; all runs and their scores are disclosed here.
- KD uses only teacher predictions on training-split windows — no external data, no test exposure; the teachers were trained exclusively on the supplied text and selected on validation.
- `common.py`, `evaluate.py`, data and tokenizer are unchanged (SHA-256 recorded in every score JSON); the contract tests pass 7/7 for the student implementation.
- Reused code: KellerJordan's `muon.py` (MIT). Substantive AI assistance (Anthropic Claude Code) was used for implementation scaffolding, training-recipe parameterization, ensemble/averaging tooling, experiment orchestration, the CPU inference fast path and drafting assistance for this report; all experimental design decisions, hyperparameter choices, result analyses and final text were reviewed and owned by the author. The 2026-09-29 development additionally used OpenAI Codex for checkpoint inspection, calibration runs and experiment documentation.

## 5 Reproduction

Full reproduction instructions — environment, the frozen-evaluation command and the complete training chain — are in `README.md`. The final submission is a single checkpoint `code/runs/p2-avg2-2400/calib-grid/best.pt` (10.55M params, single forward pass at scoring, $T = 1.08$), SHA-256 `aca6579fc009ca3e75b8a406249dc175004a29b6e1a93fb0cf9512db10127608`, distributed as the checkpoint bundle linked from the course submission form. Expected frozen score: test BPB **1.414029** in 31.61 s CPU FP32 (4.14× the same-session baseline).

## References


[1] K. Jordan. *Muon: An optimizer for hidden layers in neural networks.* `https://github.com/KellerJordan/Muon`, 2024.

[2] N. Vyas et al. SOAP: Improving and Stabilizing Shampoo using Adam. ICLR 2025.

[3] K. Wen et al. Fantastic Pretraining Optimizers and Where to Find Them. arXiv:2509.02046, 2025.

[4] I. Loshchilov and F. Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts. ICLR 2017.

[5] E. Grave, A. Joulin, N. Usunier. Improving Neural Language Models with a Continuous Cache. ICLR 2017.

[6] S. Ainsworth, J. Hayase, S. Srinivasa. Git Re-Basin: Merging Models modulo Permutation Symmetries. ICLR 2023.

[7] S. Merity et al. Pointer Sentinel Mixture Models. ICLR 2017.

[8] KellerJordan et al. modded-nanogpt speedrun records (backout lambda). `https://github.com/KellerJordan/modded-nanogpt`, 2025.

[9] G. Hinton, O. Vinyals, J. Dean. Distilling the Knowledge in a Neural Network. arXiv:1503.02531, 2015.

[10] T. Furlanello, Z. Lipton, M. Tschannen, L. Itti, A. Anandkumar. Born-Again Neural Networks. ICML 2018.

[11] S. Hu, Y. Tu, X. Han, et al. MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies. COLM 2024, arXiv:2404.04125.